
Feature Selection for High-Dimensional Neural Decoding: A Comparative Evaluation of Sparsity, Accuracy, and Runtime

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Abstract

We study feature selection in a high-dimensional neural decoding task with 11,190 neuron–timebin features and 683 trials. The dataset exhibits extreme $p \gg n$ imbalance ($p/n \approx 16$) and class imbalance (Left: 30.5%, Right: 69.5%), making stable generalisation difficult. We compare six feature selection paradigms: Forward Stepwise Selection (FSS), mRMR, Lasso, Elastic Net, Random Forest VI, and Gradient Boosting VI within a unified evaluation pipeline that ranks features, evaluates performance across subset sizes K , and assesses generalisation at optimal K^* . FSS yields the best overall trade-off, achieving strong performance (BalAcc = 0.774) with compact subsets ($K^* = 30$). Gradient Boosting VI performs similarly (BalAcc = 0.729) with slightly larger subset ($K^* = 35$), while Elastic Net achieves the highest BalAcc (0.805) at the cost of dense, less transferable solutions ($K^* = 2168$). We discuss why these methods behave differently and outline limitations affecting reliability of our findings.

1 Introduction

High-dimensional neural decoding is challenging because behavioural information is often weakly distributed across thousands of noisy neural features, producing extremely low marginal signal-to-noise. Classical assumptions of sparsity or dominant predictors rarely hold, and spike-count features introduce additional non-Gaussian noise and instability (Paninski, 2004). This motivates our central question:

Which feature selection method is most effective when information is globally distributed and each feature exhibits low signal-to-noise?

We study six feature selection methods spanning three families:

- **Wrapper:** Forward Stepwise Selection (FSS), Minimum Redundancy Maximum Relevance (mRMR)
- **Shrinkage-based:** Lasso, Elastic Net
- **Tree-based:** Random Forest Variable Importance, Gradient Boosting Variable Importance

These approaches capture different methods: linearity and sparsity (shrinkage), greedy joint modelling (wrapper), and non-linear interaction learning (trees).

2 Related Work

Feature selection in high dimensions has been extensively studied. Shrinkage-based methods; Lasso (Tibshirani, 1996) and Elastic Net (Zou & Hastie, 2005) impose convex penalties that encourage sparsity. Lasso struggles when features are highly correlated, arbitrarily selecting one from correlated groups, thus not well-suited for $p \gg n$ settings. Elastic Net addresses this by combining ℓ_1 and ℓ_2 penalties, encouraging grouping effects where correlated predictors enter or exit together, thus performs well when $p \gg n$. Stepwise procedures greedily explore subsets and mRMR (Peng et al., 2005) minimises redundancy while maximising relevance. Tree-based approaches i.e., Random Forests (Breiman, 2001) and Gradient Boosting (Friedman, 2001) extract feature importance from

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ensemble splits. RF importances can degrade in ultra-high dimensions due to individual tree overfitting, whereas Gradient Boosting’s stagewise additive approach and residual-based fitting may better capture weak, distributed signals. Neural decoding adds complexity: spike-count features exhibit non-Gaussian statistics and distributed weak-signal patterns (Paninski, 2004). Prior work emphasises need for regularisation and stability in population coding models. Our work analyses how feature selection paradigms behave under these structural constraints through a unified protocol.

3 Problem Setup and Notation

The dataset contains spike-count recordings from mouse frontal cortex during a binary decision-making task (IBL et al., 2023), with $p = 11,190$ neuron–timebin features across $n = 683$ trials. We denote each trial by index $i = 1, \dots, n$, with feature vector $x_i \in \mathbb{R}^p$ and binary behavioral label $y_i \in \{0, 1\}$. Let $X \in \mathbb{R}^{n \times p}$ denote the feature matrix and $y \in \{0, 1\}^n$ the vector of labels.

We evaluate model performance using **balanced accuracy**, which weights both classes symmetrically to account for class imbalance (30.5% / 69.5%):

$$\text{BA} = \frac{1}{2} \left(\frac{\text{TP}}{\text{TP} + \text{FN}} + \frac{\text{TN}}{\text{TN} + \text{FP}} \right).$$

We then construct **feature subsets** S_K using two approaches:

- *Ranking-based methods* (FSS, mRMR, RF-VI, GB-VI) produce an ordered ranking $r(1), \dots, r(p)$, where $r(k)$ denotes the k -th most informative feature. For any $K \leq p$, we define $S_K = \{r(1), \dots, r(K)\}$ and $X_{S_K} = X(:, S_K)$.
- *Shrinkage-based methods* (LASSO, Elastic Net) fit penalised logistic regression with parameters λ (LASSO) or (λ, α) (Elastic Net). For each regularisation parameter setting, the fitted model selects a subset S_K of features with non-zero coefficients. We vary penalty strength to obtain subsets of different sizes $K = |S_K|$.

Finally, we measure **runtime** as total time for: feature selection and evaluating balanced accuracy across all subset sizes K on the test set.

4 Evaluation Pipeline

Step 1: Optimal subset selection: For ranking-based methods, we evaluate subsets $K = 5, 10, 15, \dots, 200$ using method-specific classifiers: logistic regression for FSS and mRMR, Random Forest for RF-VI, and Gradient Boosting for GB-VI. We define optimal subset size as

$$K^* = \arg \max_K \text{BA}(S_K).$$

For LASSO and Elastic Net, we perform 5-fold cross-validation across the regularisation path. For each parameter setting, we obtain a subset S_K of non-zero features and define optimal subset size as

$$K^* = \arg \max_{S_K} \text{CV-BA}(S_K),$$

where $\text{CV-BA}(S_K)$ is the cross-validated balanced accuracy for the subset of non-zero features S_K . We then fit the model with the K^* non-zero features on the training set and evaluate on the held-out test set for final balanced accuracy.

Step 2: Method ranking: We rank methods using a composite score combining K^* , balanced accuracy, and runtime. For each criterion, we rank all $n = 6$ methods: K^* and runtime in ascending order (lower is better), balanced accuracy in descending order (higher is better). Ranks are assigned using the minimum method (ties receive the same rank).

Each rank is then converted to a score out of 100:

$$\text{Score} = \frac{n + 1 - \text{Rank}}{n} \times 100.$$

Individual scores are then weighted and summed ($w_K = 0.45$, $w_{\text{BA}} = 0.45$, $w_{\text{RT}} = 0.10$):

$$\text{Total Score}(m) = w_K \cdot \text{K_score}(m) + w_{\text{BA}} \cdot \text{BA_score}(m) + w_{\text{RT}} \cdot \text{RT_score}(m),$$

For example, FSS with:

- $K^* = 30 \rightarrow \text{rank } 1 \rightarrow \text{K_score}(\text{FSS}) = \frac{6+1-1}{6} \times 100 = 100$
- $\text{BalAcc} = 0.774 \rightarrow \text{rank } 2 \rightarrow \text{BA_score}(\text{FSS}) = \frac{6+1-2}{6} \times 100 = 83.33$

• Runtime = 5.38 hrs $\rightarrow$ rank 6 $\rightarrow$ $RT_score(FSS) = \frac{6+1-6}{6} \times 100 = 16.67$
Total Score(FSS) = $0.45(100) + 0.45(83.33) + 0.10(16.67) = 84.2$.

Step 3: Cross-classifier comparison: At each method's K^* , we evaluate four classifiers:

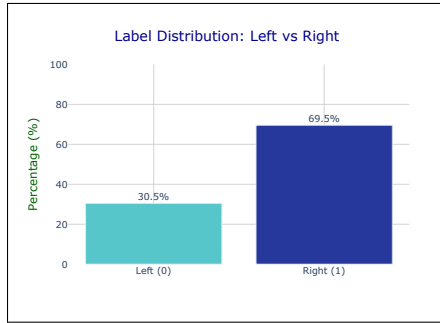
- **Logistic Regression:** solver=liblinear, class_weight=balanced
- **Linear SVM:** kernel=linear, class_weight=balanced
- **kNN:** $k = 10$
- **Random Forest:** n_estimators=100, max_depth=5, class_weight=balanced

We compute average balanced accuracy across classifiers for each feature selection method.

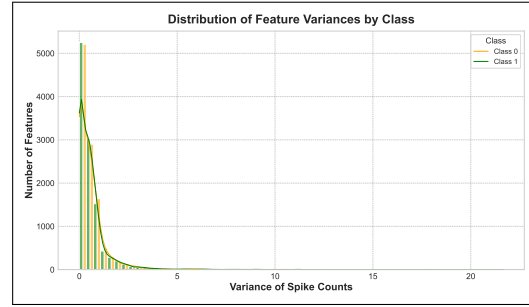
5 Exploratory Data Analysis

We extract the following insights from EDA:

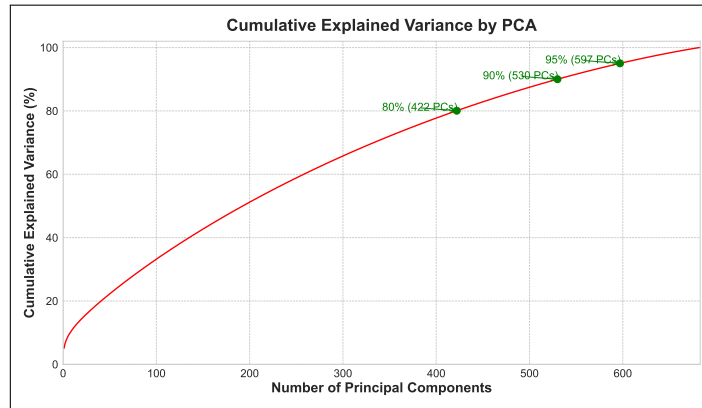
- **High $p \gg n$ setting:** With $p/n \approx 16$, standard classifiers overfit without dimensionality reduction or regularisation. Feature selection is essential for variance control and computational tractability.
- **Class imbalance:** Behavioural labels are skewed (Left: 30.5%, Right: 69.5%; Fig. 1a). Since standard accuracy is misleading, balanced accuracy is used throughout.
- **Low signal-to-noise ratio:** Multiple indicators show that predictive signal is weak but distributed across many features:
 - **Sparse signals:** Spike-count histograms show most features have ≈ 0 values (Appendix Fig. 11, 12).
 - **Low-variance features:** Variance analysis reveals 536 features with zero variance and $\approx 6,000$ have variance ≤ 0.5 (Figure 1b), suggesting many appear uninformative in isolation.
 - **Distributed information:** PCA shows no clear elbow; 80%, 90%, and 95% variance require 422, 530, and 597 components (Figure 1c).
 - **Low inter-feature correlation:** Average $|\rho| < 0.05$ (Appendix Fig. 13), indicating weak but complementary signals across features.



(a) Class distribution



(b) Marginal feature variance



(c) PCA cumulative explained variance

Figure 1: Exploratory data analysis: (a) class imbalance, (b) feature variance distribution, (c) PCA variance structure.

Together, these properties imply that no trivially identifiable feature subset exists. Removing too many predictors risks discarding real signal, while retaining too many harms generalisation. We therefore expect the optimal subset size K^* for each method to lie in a mid-range rather than being extremely small or extremely large.

6 Results

Table 1 summarises the results from our analysis. FSS achieves competitive accuracy (0.774) with a highly compact subset of features ($K^* = 30$), while GB-VI produces a slightly larger subset ($K^* = 35$) with strong performance (0.770). Elastic Net achieves the highest BalAcc (0.805) but requires a substantially larger feature subset ($K^* = 2168$), which explains its lower overall ranking despite superior predictive performance. Notably, RF-VI, mRMR, and Lasso select $K^* = 70, 185, 30$ respectively and yield lower BalAccs (0.753, 0.634, 0.729); we examine possible explanations in section 7: Discussion. Figure 2 shows the trajectory each method follows to reach its optimal BalAcc across different values of K . Tree-based methods (RF-VI, GB-VI) complete feature selection and evaluation in <10 minutes, while wrapper methods (FSS, mRMR) incur runtimes >4.5 hours. Based on the composite scoring framework that balances accuracy, sparsity, and runtime, FSS achieves the highest overall score (84.2), followed by GB-VI (74.2).

Method	K^*	BalAcc	RT (h)	K^* Score	BalAcc Score	RT Score	Total Score
FSS	30	0.774	5.383	45.0	37.5	1.7	84.2
GB-VI	35	0.770	0.145	37.5	30.0	6.7	74.2
RF-VI	70	0.753	0.039	30.0	22.5	8.3	60.8
ENet	2168	0.805	1.617	7.5	45.0	5.0	57.5
Lasso	350	0.729	0.022	15.0	15.0	10.0	40.0
mRMR	185	0.634	4.534	22.5	7.5	3.3	33.3

Table 1: Comparison of feature selection methods. Weighted scores are computed by ranking K^* and runtime in ascending order, BalAcc in descending order, converting ranks to scores out of 100, then applying weights ($w_K = 0.45$, $w_{\text{BalAcc}} = 0.45$, $w_{\text{RT}} = 0.10$).

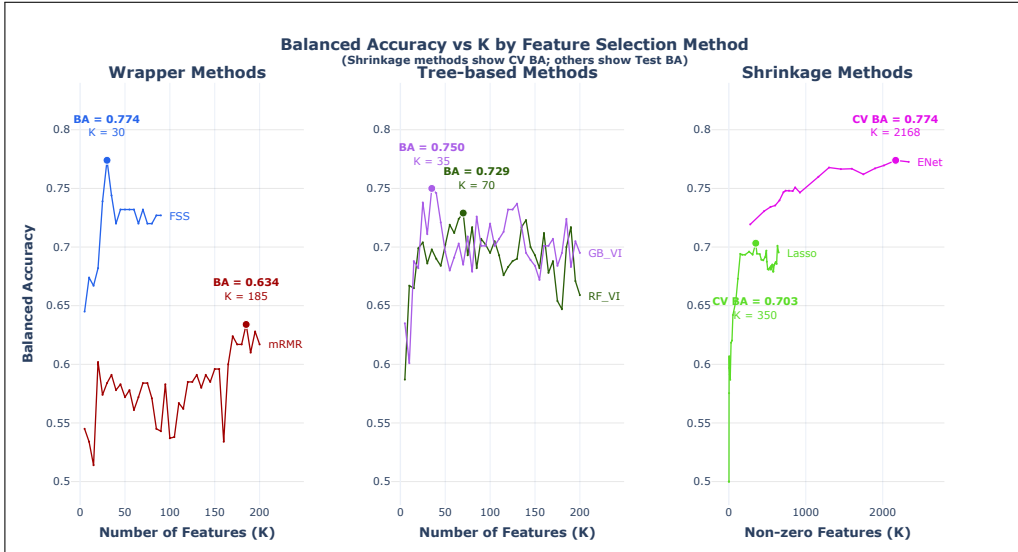


Figure 2: Balanced accuracy versus K grouped by paradigm: (left) wrapper, (center) tree-based, (right) shrinkage. FSS peaks early then plateaus; tree-based methods show similar early peaks with more variation; shrinkage methods require larger K , with Elastic Net improving up to $K^* = 2168$. Shrinkage methods display CV BA; others show test BA.

Using the optimal K^* from each method, we fitted four classifiers on the selected feature subsets. The results are summarised in Table 2 / Figure 3.

Method	K^*	LogReg	SVM _{linear}	kNN	RF	Avg BalAcc
ENet	2168	0.809	0.729	0.560	0.617	0.679
FSS	30	0.774	0.751	0.675	0.708	0.727
GB-VI	35	0.770	0.743	0.697	0.737	0.737
Lasso	350	0.699	0.707	0.627	0.676	0.677
RF-VI	70	0.699	0.703	0.641	0.753	0.699
mRMR	185	0.634	0.621	0.552	0.559	0.592

Table 2: Balanced accuracy of four classifiers trained on each method’s optimal feature subset.

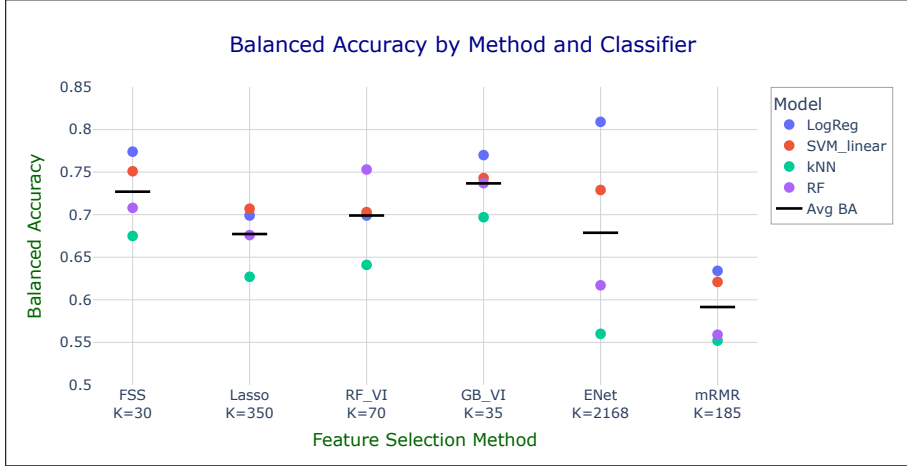


Figure 3: Balanced accuracy for four classifiers trained on optimal subsets. FSS and GB-VI achieve highest mean BalAcc across classifiers. Elastic Net performs well only on linear models but poorly on kNN and RF due to curse of dimensionality.

GB-VI and FSS achieve the highest mean BalAcc (0.737 and 0.727 respectively) across all 4 classifiers due to compact, highly informative feature subsets selected ($K^* = 35$ and $K^* = 30$), which reduce model variance and enable stable generalisation across diverse classifier types.

Elastic Net exhibits strong performance on linear models but poor generalisation to non-linear classifiers. With $K^* = 2168$, it achieves the highest individual BalAcc (0.809) on LogReg and performs well on linear SVM (0.729), but degrades substantially on kNN (0.560) and Random Forest (0.617), yielding a mean BalAcc of only 0.679.

mRMR and Lasso achieve the lowest mean BalAccs (0.592 and 0.677). Both methods struggle in this low-correlation regime (average $|\rho| < 0.05$): with minimal feature redundancy, redundancy minimisation criterion (mRMR) and ℓ_1 penalty cannot exploit correlated groups to stabilise coefficient paths, resulting in suboptimal feature selection.

7 Discussion

Forward Stepwise Selection (FSS) achieves the highest total score in our ranking (84.2) with the smallest feature subset ($K^* = 30$), high BalAcc on the test set (0.774), and strong generalisation across classifiers (mean BalAcc = 0.727). Its strong predictive performance stems from its greedy AIC-driven search: at each step, it adds the feature that minimizes AIC, which approximates expected test risk under log-loss. This joint optimization of prediction and feature selection explains why FSS discovers the "first layer" of neural features with strongest marginal contributions; AIC decreases sharply then plateaus around $K = 30$ (Fig. ??), effectively isolating the most signal-dense features. *However, FSS has limitations:* it becomes computationally prohibitive (runtime >5 hrs), and the plateau indicates that once the strongest marginal features are selected, the greedy search becomes unstable and fails to discover additional weak but collectively informative features. This makes FSS ideal for finding a minimal feature subset for interpretability but limited when higher BalAcc is the priority.

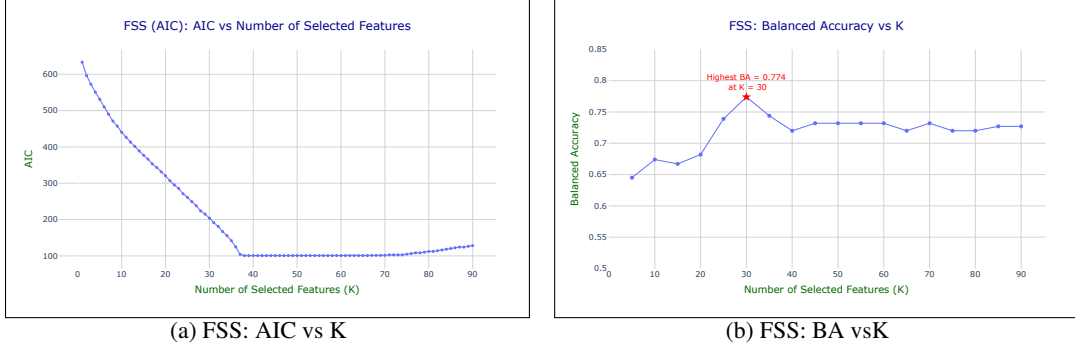


Figure 4: (a) FSS: AIC vs K . Plateaus around $K \approx 35$ (b) FSS: BalAcc vs K . Balanced accuracy steadily rises, peaks at $K = 30$ and plateaus thereafter. FSS achieves second highest BalAcc and generalises well among diverse classifier types

Gradient Boosting Variable Importance (GB-VI) achieves the second-best total score (74.2): compact subset ($K^* = 35$), high test BalAcc (0.770), and highest avg BalAcc across classifiers (0.737). Unlike RF, which builds deep trees in parallel and averages predictions, GB builds shallow trees sequentially, fitting each to the residuals of the previous ensemble. This concentrates importance on genuinely informative features: GB achieves 80% cumulative importance with only 143 features (Fig. 5a) vs 796 for RF (Fig. 6a). This accumulation of weak, local improvements aligns well with distributed weak-signal datasets where no single feature is strongly predictive but many small increments collectively matter. Runtime is highly efficient (<10 minutes), making GB-VI a strong competitor to FSS when both sparsity and balanced accuracy are priorities.

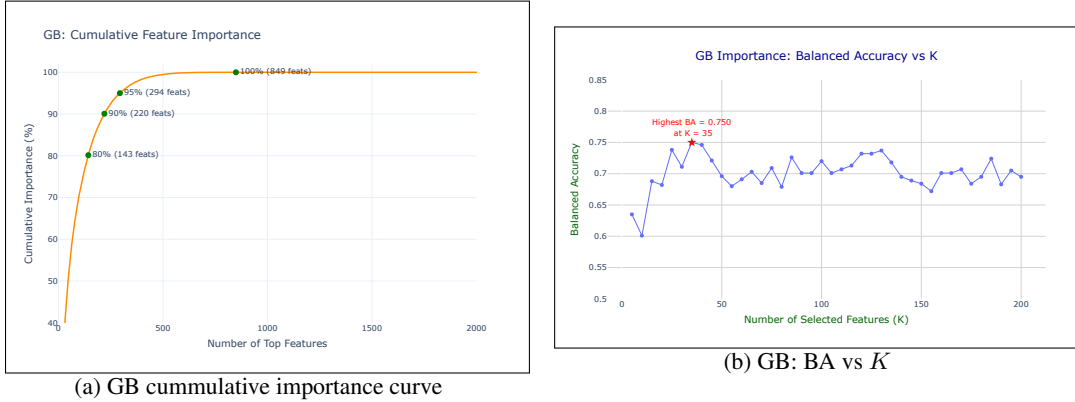
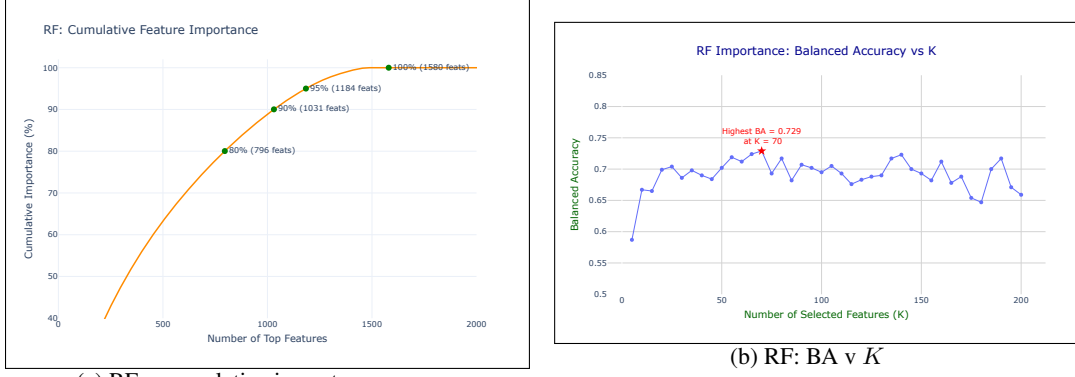


Figure 5: (a) GB requires only 143 feats to achieve $\approx 80\%$ cumulative importance compared to 796 for RF (b) GB achieves peak BalAcc at 35 features demonstrating superior concentration on informative features

Random Forest Variable Importance (RF-VI) achieves modest performance ($K^* = 70$, BalAcc = 0.753) but low mean BalAcc across classifiers (0.699). Two issues limit this method. First, impurity-based splits are biased toward high-variance features, which can be noisy rather than informative. In our dataset, where most features have low variance ($\sim 6,000$ features with variance ≤ 0.5), this bias causes RF-VI to favour noisy, high-variance features over weakly informative predictors. The cumulative feature importance curve confirms this: RF requires 796 features to reach 80% cumulative importance (Fig. 6a). Second, in our $p \gg n$ setting, each tree overfits severely with terminal nodes containing very few observations. Although ensemble averaging reduces variance, aggregated importance rankings are unreliable as individual trees are fundamentally unstable.



(a) RF cumulative importance curve
Figure 6: (a) RF requires 796 features for $\approx 80\%$ cumulative importance (b) To reach peak BalAcc, RF requires a higher feature subset of $K = 70$

Elastic Net achieves the highest single-classifier BalAcc (0.805 on LogReg) by retaining a large feature subset ($K^* = 2168$). CV selects $\alpha = 0.2$ and $C = 0.45$, (ℓ_2 -dominant). It also performs well on linear SVM (0.729), but fails to generalise to other classifiers. kNN (BalAcc = 0.560) suffers from the curse of dimensionality, and Random Forest (BalAcc = 0.617) struggles with variance instability. As a result, avg BalAcc across classifiers (0.679) is modest despite superior linear model performance. Runtime (1.6 hrs) is less than FSS and mRMR due to a coarse grid search; finer tuning would increase runtime, yet even with coarse search Elastic Net yields the highest BalAcc.

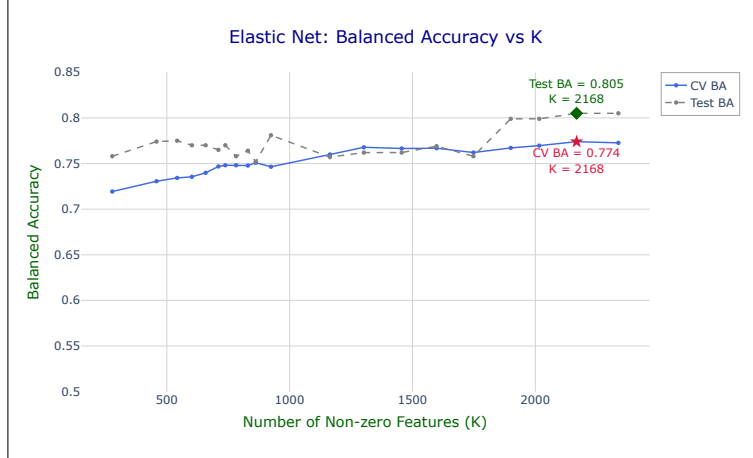


Figure 7: Elastic Net balanced accuracy vs K for different values of mixing parameter α . CV-optimal configuration ($\alpha = 0.2$, $C = 0.45$) selects $K^* = 2168$ features where both CV BA and test BA peak (test BalAcc = 0.806). Limited hyperparameter tuning contributes to this alignment.

LASSO ranks second-last with a large feature subset ($K^* = 350$) and BalAcc = 0.729. While it performs reasonably on logistic regression, it generalises poorly to other classifiers (mean BalAcc = 0.677). In typical high-correlation settings, LASSO struggles because the ℓ_1 penalty arbitrarily selects one feature from correlated groups. However, with low inter-feature correlation (average $|\rho| < 0.05$), LASSO can retain more features without redundancy penalties. Despite this, LASSO imposes hard sparsity which discards weakly informative features that collectively contribute to prediction. In a distributed weak-signal regime, this aggressive shrinkage eliminates signal prematurely, explaining why LASSO underperforms relative to Elastic Net.

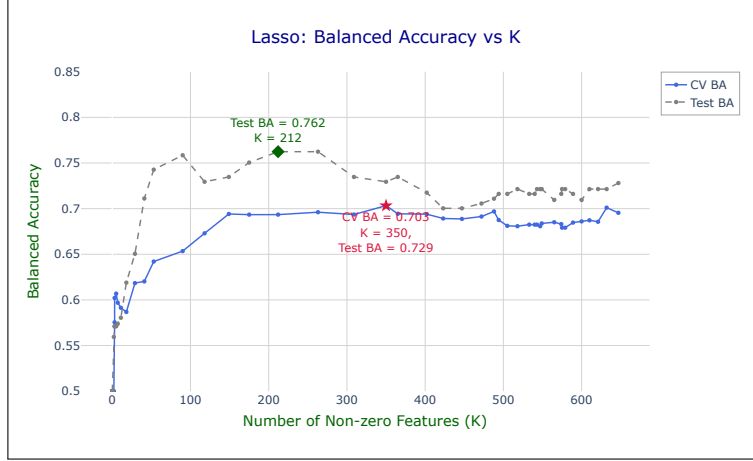


Figure 8: LASSO: Balanced accuracy vs K . CV-optimal configuration selects $K^* = 350$ features, achieving BalAcc = 0.729 on the test set. Note that test BA peaks earlier at $K = 212$ (BalAcc = 0.74), but we use CV-optimal to prevent data leakage.

Minimum Redundancy Maximum Relevance (mRMR) achieves the lowest total score (33.3) and ranks last: $K^* = 185$ features that prove least informative (BalAcc = 0.634, lowest avg BalAcc = 0.592). This poor performance stems from two structural mismatches with our dataset. First, mRMR relies on mutual information (MI) estimation between features and labels. With $p \gg n$, MI estimates are highly unstable; the top 200 ranked features show shallow decay with nearly indistinguishable MI values (Fig. 9), leading to noisy feature rankings. Second, mRMR’s core principle of minimising redundancy offers no advantage where features exhibit low mutual correlation. When features are largely independent, redundancy minimisation becomes irrelevant, and mRMR degenerates into a purely relevance-based ranking. Additionally, the computational runtime exceeds 4.5 hours, making mRMR entirely unsuitable for our dataset.

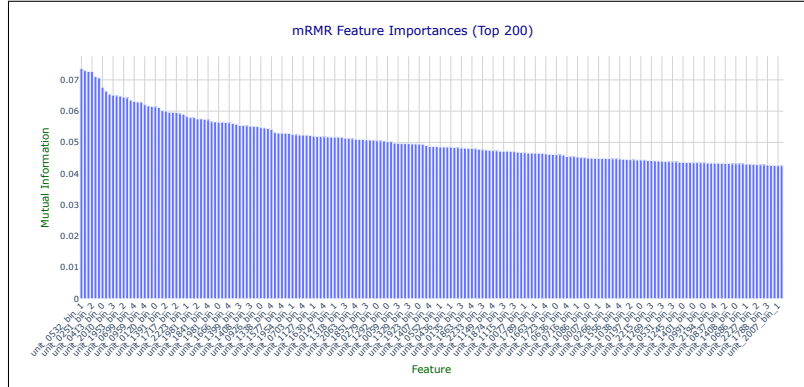


Figure 9: MI-based importance scores for top 200 mRMR features. Shallow decay reflects unstable estimation in $p \gg n$ regime.

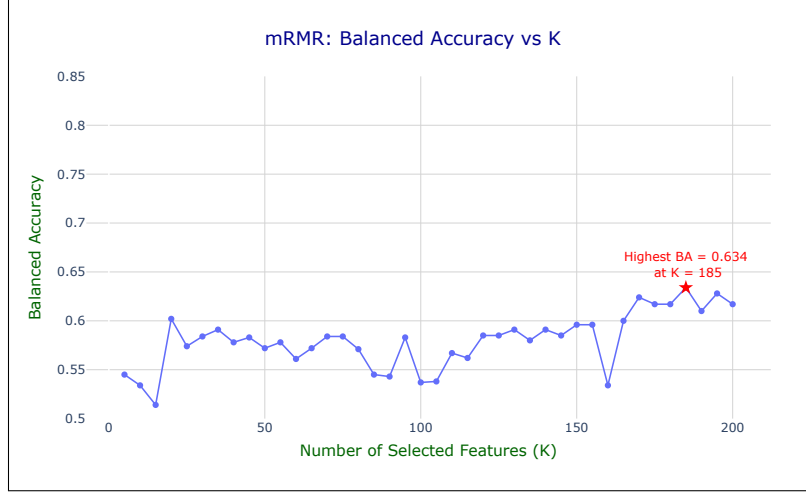


Figure 10: mRMR: Balanced accuracy vs K . BalAcc peaks at $K = 185$ (0.634), the lowest among all methods. The erratic trajectory with no clear peak reflects the unstable mutual information estimates in the $p \gg n$ regime, where feature rankings are noisy and unreliable.

7.1 Classifier Performance Across Feature Subsets

Feature selection method matters more than classifier choice. Methods fall into three groups:

Compact and highly informative (FSS: $K^* = 30$, GB-VI: $K^* = 35$): Distinguish signal from noise, producing small subsets that generalize well (avg BalAcc ≥ 0.72).

Moderately compact, mixed informativeness (RF-VI: $K^* = 70$, mRMR: $K^* = 185$): RF-VI achieves modest avg BalAcc (0.70); mRMR’s larger, uninformative subset yields lowest avg BalAcc (0.59).

Large subsets, variable informativeness (LASSO: $K^* = 350$, Elastic Net: $K^* = 2168$): LASSO’s features prove weakly informative (avg BalAcc = 0.68); Elastic Net’s dense subset excels on linear models but degrades on kNN and RF (avg BalAcc = 0.68).

8 Conclusion, Limitations, and Next Steps

We sought to identify the best feature selection method for high-dimensional neural decoding with weakly correlated features and sparse signals. Performance depends critically on how each method identifies important features in this setting. FSS and GB-VI succeed by identifying compact, highly predictive subsets that generalize well (FSS via greedy AIC minimization, GB-VI via sequential residual fitting). RF-VI underperforms due to bias toward high-variance features and tree overfitting. mRMR fails entirely as redundancy minimization offers no benefit when features are weakly correlated. Among shrinkage methods, LASSO’s hard sparsity discards weak signals prematurely, while Elastic Net retains a large informative subset achieving highest BA on linear models (0.805) but failing on distance-based classifiers. Depending on priorities—interpretability (FSS), generalization (GB-VI), or peak linear performance (Elastic Net)—practitioners can select accordingly.

However, our analysis relies on a single train-test split, coarse grid search, and basic class imbalance handling (`class_weight=balanced`), yielding a performance ceiling of BalAcc = 0.805. Further analysis can incorporate repeated cross-validation for stable performance estimates, finer hyperparameter tuning, advanced resampling techniques (SMOTE, ensemble methods), feature stability analysis via bootstrapping, and domain-aware feature engineering (temporal smoothing, neuron interactions) or non-linear models (kernel SVM, neural networks) to push beyond the current ceiling.

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Appendix

A Feature Selection Methods: Implementation Details

We evaluate six feature selection methods: wrapper (FSS, mRMR), shrinkage-based (LASSO and Elastic Net), and tree-based embedded approaches (Random Forest and Gradient Boosting Variable Importances). Below we provide the mathematical formulations for each method.

Forward Stepwise Selection is a greedy wrapper method that iteratively adds the feature that yields the largest decrease in Akaike Information Criterion (AIC). At step k , we choose

$$j_k = \arg \min_{j \notin S_{k-1}} \text{AIC}(S_{k-1} \cup \{j\}),$$

where $\text{AIC}(S)$ is computed from a logistic regression fitted on subset S . FSS does not require hyperparameter tuning. We evaluate up to $K_{\max} = 200$ features.

Minimum Redundancy Maximum Relevance (mRMR) ranks features by maximising mutual-information relevance while minimising redundancy:

$$j_k = \arg \max_{j \notin S_{k-1}} \left[I(x_j; y) - \frac{1}{|S_{k-1}|} \sum_{\ell \in S_{k-1}} I(x_j; x_\ell) \right].$$

mRMR is deterministic and ranks up to $K_{\max} = 1500$ features.

LASSO performs feature selection via ℓ_1 -penalised logistic regression,

$$\hat{\beta} = \arg \min_{\beta} \left(-\ell(\beta; X, y) + \lambda \|\beta\|_1 \right).$$

We tune $C = 1/\lambda$ via 5-fold cross-validation over 50 logarithmic values in $[10^{-3}, 10^1]$.

Elastic Net combines ℓ_1 and ℓ_2 penalties:

$$\hat{\beta} = \arg \min_{\beta} \left(-\ell(\beta; X, y) + \lambda_1 \|\beta\|_1 + \lambda_2 \|\beta\|_2^2 \right).$$

We tune $\alpha \in \{0.2, 0.8\}$ and $C \in \{0.05, 0.10, \dots, 0.50\}$ via 5-fold cross-validation.

Random Forest Variable Importance (RF-VI) ranks features using impurity reduction across trees:

$$\text{VI}_{\text{RF}}(j) = \sum_{t=1}^T \sum_{\text{splits } s \text{ on } j} \Delta \text{Gini}(s).$$

We tune via 5-fold CV: `n_estimators` $\in \{100, 200, 300, 400\}$, `max_depth` $\in \{5, 10, 20, \text{None}\}$, `min_samples_leaf` $\in \{1, 2, 5\}$, `max_features` $\in \{\text{sqrt}, \log 2\}$.

Gradient Boosting Variable Importance (GB-VI) computes importance from the total loss reduction contributed by feature j :

$$\text{VI}_{\text{GB}}(j) = \sum_{t=1}^M \sum_{\text{splits } s \text{ on } j} \Delta \mathcal{L}_t(s).$$

We tune via 5-fold CV: `n_estimators` $\in \{100, 200\}$, `learning_rate` $\in \{0.05, 0.1\}$, `max_depth` $\in \{3, 4\}$, `subsample` $\in \{0.8, 1.0\}$.

B Additional EDA

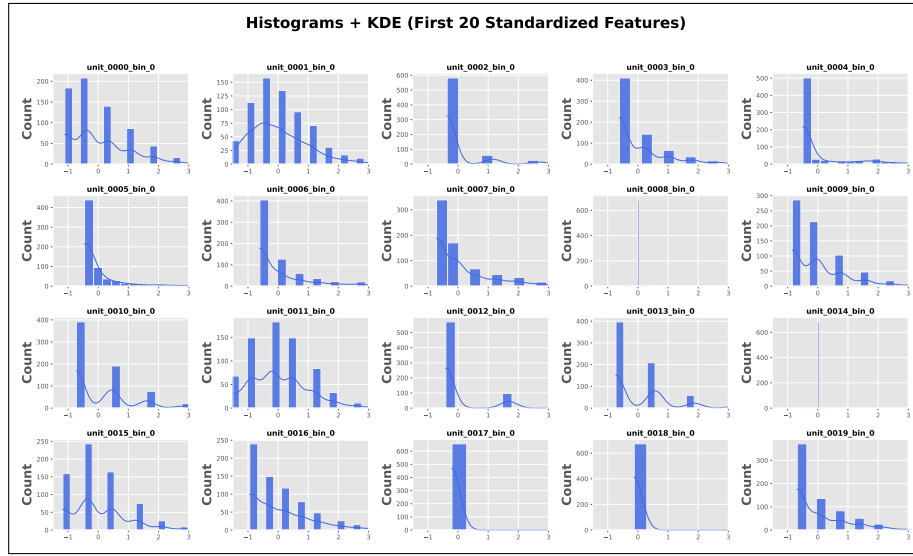


Figure 11: Histograms with KDE overlays for the first 25 standardised features. Most features exhibit right-skewed distributions with a dominant mode near zero, reflecting sparse neural firing patterns typical of spike-count data.

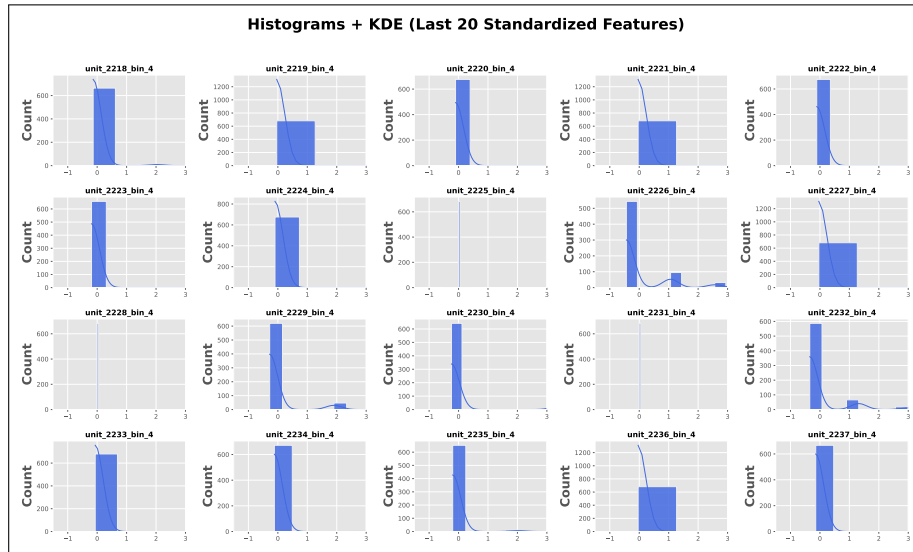


Figure 12: Histograms with KDE overlays for the last 25 standardised features. These later time-bin features show even stronger zero-inflation and reduced variance compared to earlier bins, suggesting weaker signal in later temporal windows.

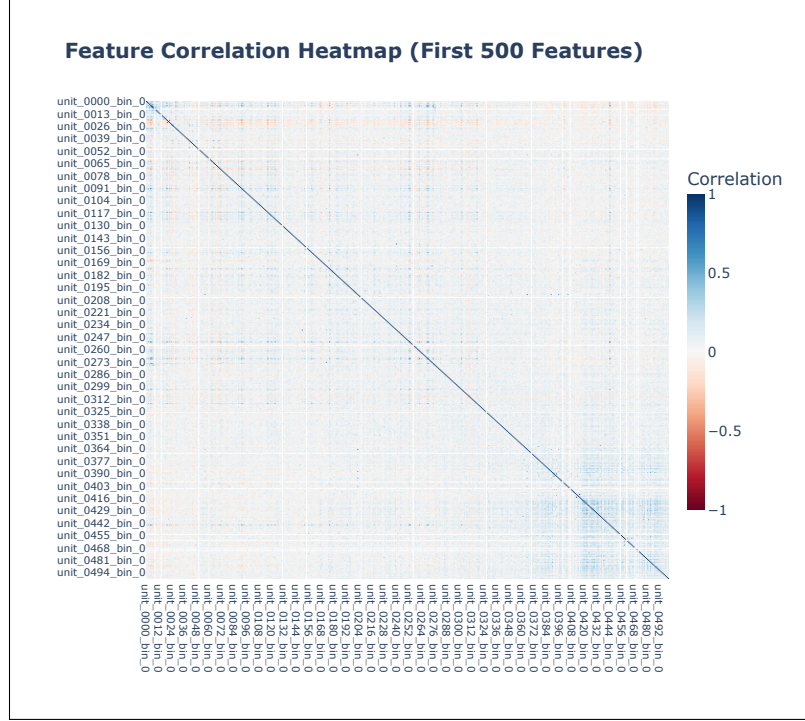


Figure 13: Correlation matrix of the first 500 features. Off-diagonal values are uniformly low (average $|\rho| = 0.046$), indicating weak inter-feature dependence. This low correlation structure explains why LASSO retains more features ($K^* = 212$) than typically expected and why mRMR’s redundancy minimisation offers little benefit.

C Feature Selection Performance

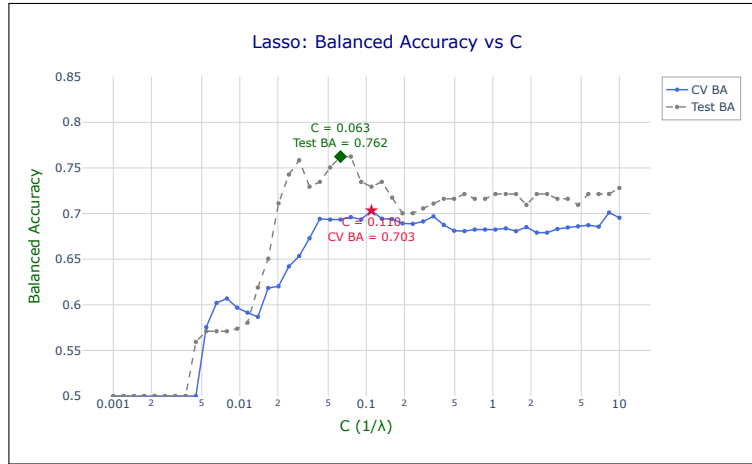


Figure 14: LASSO: Balanced accuracy vs regularisation parameter C . CV-optimal configuration at $C = 0.110$ (corresponding to $K^* = 350$ non-zero features).

D Feature Selection Evaluation

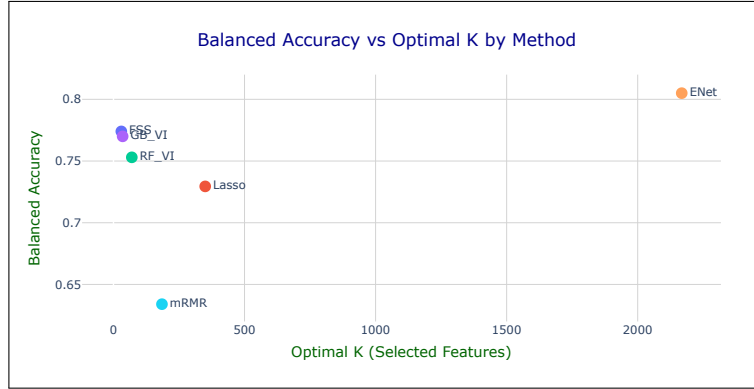


Figure 15: Balanced accuracy versus optimal subset size K^* for all methods. FSS and GB-VI achieve high BalAcc with compact subsets ($K^* < 40$), while Elastic Net requires over 2000 features to attain the highest BalAcc.

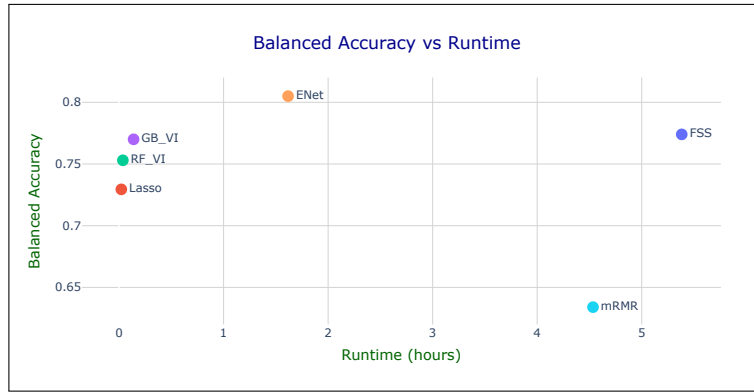


Figure 16: Balanced accuracy versus runtime (hrs). Tree-based methods (RF-VI, GB-VI) and LASSO complete in under 10 minutes, while wrapper methods (FSS, mRMR) require over 4 hours due to iterative model refitting.



Figure 17: (Left) Weighted scores by criterion: balanced accuracy (45%), sparsity/ K^* (45%), and runtime (10%). (Right) Overall composite scores showing FSS ranks highest (84.2) due to excellent sparsity ($K^* = 30$) and strong balanced accuracy (0.774), while mRMR ranks lowest (33.3) due to poor balanced accuracy (0.634) despite moderate sparsity. The composite framework reflects the priorities of interpretability and prediction quality.

E Feature Overlap Analysis

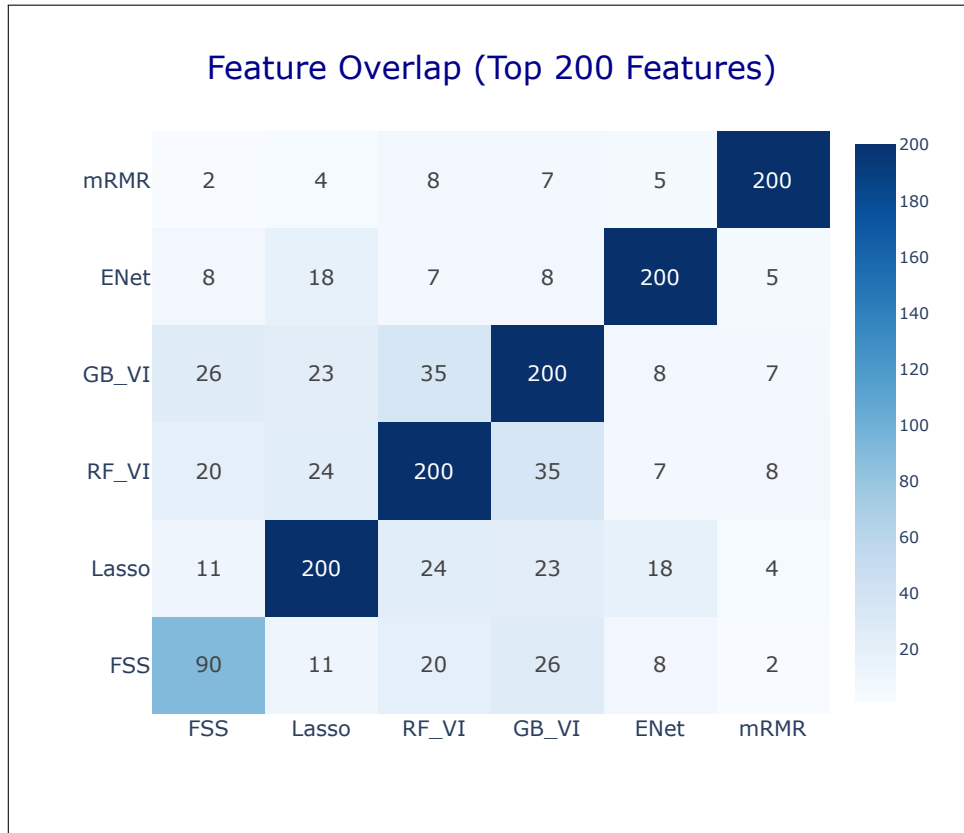


Figure 18: Overlap of the top 200 selected features across all six methods. Low pairwise agreement (maximum 35 shared features between any two methods) reflects the distributed neural encoding structure with no dominant predictors.